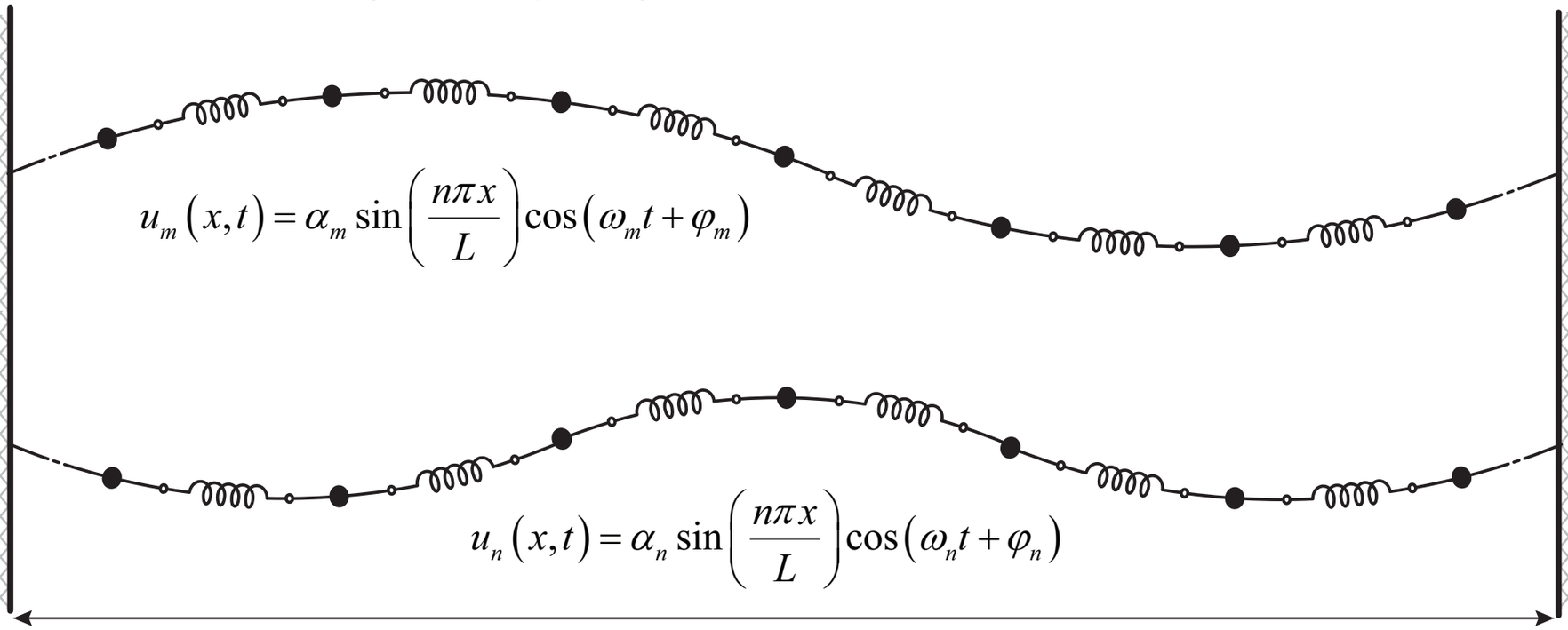
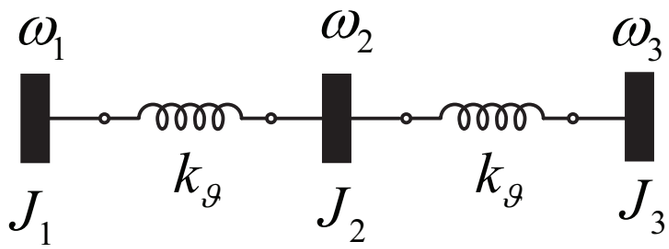
two dominant linear dispacement harmonics

$$\frac{\partial^2 u(x,t)}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 u(x,t)}{\partial t^2} \longrightarrow u_m(x,t) + u_n(x,t)$$



isomorphism (from PDEs to ODEs)

two dominant torsional displacement harmonics



$$\begin{cases} J_1\dot{\omega}_1 = -\tau_{12} \\ J_2\dot{\omega}_2 = \tau_{12} - \tau_{23} \\ J_3\dot{\omega}_3 = \tau_{23} \\ \dot{\tau}_{12} = k_\vartheta(\omega_1 - \omega_2) \\ \dot{\tau}_{23} = k_\vartheta(\omega_2 - \omega_3) \end{cases}$$